

BSc (Hons) in Information Technology Specialized in AI IT3012 : Intelligent Agents Year 3 - Semester 1 - 2026 Semester 2

Objective & Lecture Mapping: Extend the search agent with Manhattan and Euclidean heuristics and use A* to reduce unnecessary node expansion.

Part 1: Practical Implementation

- Added `manhattan_distance()` and `euclidean_distance()`.
- Implemented A* with $f(n) = g(n) + h(n)$ and best-known path costs.
- Connected A* to the visual game. Checkpoint: (0,0) to (3,4) gives Manhattan 7 and Euclidean 5.0.

Part 2: Analytical Evaluation - Completed Answers

1. How does UCS differ from A*?

Answer: UCS prioritizes the smallest accumulated path cost $g(n)$. A* prioritizes $g(n) + h(n)$, adding an estimate of the remaining cost to guide exploration toward the goal.

2. Why is Manhattan distance admissible, and what happens if a heuristic is not?

Answer: On a four-direction unit-cost grid, any route must cover at least the horizontal and vertical separation. Manhattan distance therefore never overestimates the true cost. A non-admissible heuristic can cause A* to return a suboptimal path.

3. Which heuristic suits diagonal movement?

Answer: When a unit-cost diagonal move is allowed, Manhattan distance can overestimate. Chebyshev distance is suitable for equal-cost diagonal moves; Euclidean distance is suitable when diagonal cost follows geometric distance.

4. Suggest a stronger heuristic for collecting multiple food items.

Answer: Use the distance from the agent to the nearest food plus the minimum spanning tree cost over all remaining food. This lower bound estimates the work required to connect every remaining target.

Completion and Verification

The implementation is stored in the matching `Week_04` branch. Automated tests pass and the runnable visual environment remains available in `visual_grid_game.py`.